David Jonatan Zapeta Chavez

Senior Backend, Cloud & AI Engineer

Node.js · NestJS · Python · AWS · Event-driven architecture · Generative AI (RAG)

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PROFILE

Backend and cloud engineer with 5+ years building production systems for banking, insurance, retail and digital platforms. I specialize in APIs with Node.js (NestJS) and Python, event-driven architectures on AWS and generative AI integration. I have built payments with bank reconciliation, critical third-party integrations and the migration of a backend into domain services, always treating security, cost and operations as part of the design. I document decisions and support the team so the system can be maintained and keep growing.

KEY ACHIEVEMENTS

- In-house payment gateway alongside Stripe, with a ledger, idempotency, signed webhooks and bank reconciliation.
- Security built into development: OAuth2/JWT, least privilege, secrets management and CI/CD that blocks vulnerable deployments.
- Domain-by-domain migration to NestJS services with DDD on ECS Fargate and Kubernetes, without a full rewrite.
- Five-product insurance platform with real-time quoting for Liverpool, extensible without touching existing code.

PROFESSIONAL EXPERIENCE

Backend Developer / Cloud & DevOps · PastPost — Jul 2024 – Oct 2026

Digital legacy platform: web (Next.js + Auth0) and mobile app (Flutter + Amazon Cognito) on a shared backend.

- Designed the **serverless, event-driven backend on AWS**: Lambda and API Gateway for on-demand work; SQS, SNS and EventBridge to decouple domains; Step Functions for signing and charging flows with compensations; Kafka and RabbitMQ where retention or advanced routing was needed.
- Integrated **Seguridata digital signatures and NOM-151 certificates** behind our own interface, with queues, exponential-backoff retries and a review queue, so a provider failure never blocks the user.
- Built an **in-house payment gateway** that coexists with Stripe behind one interface: double-entry ledger, per-payment state machine, idempotency keys, outbox, HMAC-signed webhooks and reconciliation against the bank.
- Led the **domain-by-domain migration** (users, products, payments) to NestJS services using DDD and the strangler pattern: gradual traffic shifting from API Gateway, an anti-corruption layer and per-service data ownership. Always-on services run on ECS Fargate and the rest on Kubernetes.
- Unified web (Auth0) and mobile (Cognito) identity in a single JWT validation layer, with least privilege, KMS encryption and Secrets Manager.
- Implemented **RAG** with LangChain and LangGraph on the OpenAI API, with guardrails and evaluations before every release, and anchored policy fingerprints on Ethereum (Solidity) to make them verifiable.
- Infrastructure as code with Terraform, CI/CD that blocks vulnerable deployments, and observability with OpenTelemetry, CloudWatch, Grafana and Datadog. Decisions documented as ADRs, code reviews and mentoring.

Stack: NestJS, TypeScript, Python, AWS (Lambda, ECS Fargate, EC2, SQS/SNS, EventBridge, Step Functions), Kubernetes, Kafka, RabbitMQ, Terraform, Stripe, LangChain.

Full-Stack / Backend Engineer · Promass · Remote — Oct 2021 – Jul 2025

Insurance and e-commerce platforms for retail clients: Liverpool and Suburbia.

- **Mi Seguro Liverpool:** platform for five insurance products (auto, major medical, pet, PIF and home) with real-time quoting. I modeled each product as a strategy behind a common interface, so adding one does not require changing the others. React and Vite SPA with Node.js and PostgreSQL microservices.

- **Liverpool.com.mx:** Next.js and React interfaces on micro-frontends, headless content in Contentstack, delivery through the Akamai CDN and Dynatrace monitoring.
- **Suburbia insurance portal** for customers and agents: Next.js with Material UI, Auth0 with roles, a Node.js backend with PostgreSQL and adapters to existing systems.
- Led the team's backend: REST and GraphQL APIs, data modeling and ORMs, automated testing, GitLab CI/CD pipelines with Docker, and resilience with dead-letter queues (DLQ) and circuit breakers.

Stack: Node.js, React, Next.js, Vue.js, PHP, GraphQL, PostgreSQL, SQL Server, Auth0, GitLab CI/CD, Docker, Dynatrace.

Data Engineer · Banco Azteca · State of Mexico

Jan 2018 – Dec 2020

- Optimized complex queries, stored procedures and triggers behind digital banking transactions.
- Designed relational and NoSQL data models for metrics and telemetry under financial-sector security standards.
- Automated data migrations, scheduled backups and disaster recovery plans with Bash and Python.

Stack: PostgreSQL, SQL Server, MySQL, MongoDB, Python, Bash.

TECHNICAL SKILLS

Backend	Node.js, NestJS, TypeScript, Python (FastAPI), GraphQL, REST, OpenAPI
Cloud & DevOps	AWS (Lambda, API Gateway, ECS Fargate, EC2, SQS/SNS, EventBridge, Step Functions, Cognito, KMS, CloudWatch), Kubernetes, Docker, Terraform, GitHub Actions, GitLab CI/CD, Azure
Architecture	Event-driven, DDD, hexagonal, microservices, Kafka, RabbitMQ, outbox, sagas, idempotency
Generative AI	RAG, LangChain, LangGraph, Semantic Kernel, OpenAI API, embeddings, guardrails, evaluations
Data	PostgreSQL, SQL Server, MySQL, MongoDB, Redis
Security	OAuth2/OIDC, JWT, Auth0, OWASP, static analysis (SAST), secrets management, threat modeling
Frontend & Web3	React, Next.js, Vue.js · Ethereum, Solidity, smart contracts
Practices	ADRs, Spec-Driven Development, TDD, code review, mentoring, continuous delivery

EDUCATION & CERTIFICATIONS

Cybersecurity · Universidad Anáhuac · 2025

Full Stack Developer · Academlo · 2019

High school diploma · 2010 – 2013

LANGUAGES

Spanish · native

English · professional working proficiency (B2)

Case studies with architecture and decisions at jonatanchz.netlify.app/en